LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 2 trial 4 seed 14085

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running an explainability benchmarking study for compound bioactivity prediction. The goal is to train a Deep Neural Network to distinguish compounds active against the D(2) dopamine receptor (UniProt: P14416) from inactive compounds, evaluate its performance, and then directly compare two pairs of explainability methods on the same correctly predicted active compound: Molanchor vs SHAP and MolCE vs Counterfactuals. Summarize where the methods agree on the structural determinants of the predictions and where they disagree.

Assistant

I'll help you conduct this explainability benchmarking study for D(2) dopamine receptor bioactivity prediction! 🧪 Let's start by finding a relevant dataset and setting up the workflow.

(Calling tool `find_datasets`)

Assistant

Excellent! I found the D(2) dopamine receptor dataset (`P14416`). Let me load it and start the workflow. 🚀

(Calling tool `load_dataset`)

Assistant

(Calling tool `load_dataset`)

Assistant

(Calling tool `load_dataset`)

Assistant

📊 Dataset loaded successfully! 8,348 compounds with SMILES and class labels. Now let me compute ECFP fingerprints for the DNN model.

(Calling tool `compute_features`)

Assistant

🔬 Features computed (2048-bit ECFP fingerprints). Now splitting the dataset with stratification.

(Calling tool `split_dataset`)

Assistant

📊 Dataset split successfully! Now let me train the Deep Neural Network model.

(Calling tool `train_model`)

Assistant

🧠 DNN training started! Let me poll for completion.

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

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Assistant

🎉 DNN training completed with excellent performance!

Metric	Test Score
Accuracy	98.64%
AUC	0.996
Balanced Accuracy	0.986
F1 Score	0.986
Precision	0.985
Recall	0.988

Now let me export predictions on the test set.

(Calling tool `export_predictions`)

Assistant

📊 Predictions exported! Now let me select a correctly predicted active compound for explainability analysis.

(Calling tool `select_compound_for_xai`)

Assistant

🎯 Selected active compound: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O` (confidence: 1.0)

Now let me run **MolAnchor** explanation:

(Calling tool `explain_with_molanchor`)

Assistant

🔍 MolAnchor identified an anchor fragment with **precision = 1.0** and **coverage = 0.53**.

Now let me run **SHAP** explanation on the same compound:

(Calling tool `explain_smiles_with_shap`)

Assistant

📊 SHAP values computed (SHAP sum = 53.69). Let me visualize the SHAP heatmap:

(Calling tool `plot_shap_mol`)

Assistant

🔬 SHAP heatmap generated. Now let me get the top SHAP bit environments:

(Calling tool `get_top_k_bit_environments_from_shap`)

Assistant

📊 Top SHAP fragments identified. Now let me run **MolCE** explanation:

(Calling tool `explain_with_molce`)

Assistant

🔬 MolCE analysis completed. Now let me generate **Counterfactuals** for the same compound:

(Calling tool `generate_counterfactuals`)

Assistant

🎯 Counterfactuals generated! Let me visualize them:

(Calling tool `visualize_counterfactuals`)

Assistant

Let me also display the MolAnchor visualization:

(Calling tool `show_plot`)

Assistant

(Calling tool `show_plot`)

Assistant

(Calling tool `show_plot`)

Assistant

Now let me compile the comprehensive explainability benchmarking summary. 📊

🧪 Explainability Benchmarking Study: D(2) Dopamine Receptor Bioactivity Prediction

1. Model Performance Summary

Metric	Test Score
Accuracy	98.64%
AUC	0.996
Balanced Accuracy	0.986
F1 Score	0.986
Precision	0.985
Recall	0.988

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2. Compound Analyzed

SMILES: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`

- **True Label:** Active (Class 1)
- **Predicted Label:** Active (Class 1)
- **Prediction Confidence:** 1.0

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3. Pair 1: MolAnchor vs SHAP Comparison

📍 MolAnchor Results

(Calling tool `show_plot`)

Assistant

📊 SHAP Results

(Calling tool `show_plot`)

Assistant

4. Pair 2: MolCE vs Counterfactuals Comparison

🔬 MolCE Results

(Calling tool `show_plot`)

Assistant

🎯 Counterfactuals Results

(Calling tool `show_plot`)

Assistant

📊 COMPREHENSIVE EXPLAINABILITY BENCHMARKING SUMMARY

🔍 Pair 1: MolAnchor vs SHAP — Structural Determinants

WHERE THEY AGREE:

Feature	MolAnchor	SHAP	Interpretation
Diazepane Core	Anchor fragment `[*]N1CCC2(CC1)C(=O)N(C)CN2[*]`	Top SHAP bits (1280, 1791)	Both identify the **central diazepane ring system** as critical
Benzylamine Moiety	Coverage: 53%	Bit 1792: `cc(CN(C)C)c` (contribution: 5.74)	Both highlight the **benzyl-methylamine** substructure
Thienyl Group	Part of anchor	Bit 885: `cc(c)s` (contribution: 5.39)	Both identify the **bithienyl moiety** as important

WHERE THEY DISAGREE:

Aspect	MolAnchor	SHAP	Explanation
Granularity	Single coarse fragment (precision=1.0)	Multiple fine-grained bits	MolAnchor gives a **coarse** view; SHAP reveals **fine-grained** contributions
Negative Features	Not reported	Bit 552: `ccs` (contribution: **-5.37**)	SHAP identifies a **detrimental** thienyl bit; MolAnchor only shows positive anchors
Coverage	53% of molecule	Distributed across 5 bits	MolAnchor covers less; SHAP captures **diffuse** feature importance

Key Insight:

> MolAnchor provides a **high-level anchor** (the diazepane core) that captures the essential scaffold, while SHAP reveals **nuanced contributions** including both positive (benzylamine, thienyl) and negative (specific thienyl bits) influences. **They agree on the core scaffold's importance but differ in granularity and negative feature detection.**

🔬 Pair 2: MolCE vs Counterfactuals — Structural Determinants

WHERE THEY AGREE:

Finding	MolCE	Counterfactuals	Interpretation
Substituent Sensitivity	R-group site 1 is critical (contrast score = 1.0)	All 10 CFs modify the same substituent position	Both confirm the **substituent on the diazepane nitrogen** is the key determinant
Activity Reversal	R-group swaps shift prediction toward inactive	All CFs flip prediction to inactive (Class 0)	Both confirm substituent changes **reverse bioactivity**

Finding	MolCE	Counterfactuals	Interpretation
Tolerance to Change	High contrast scores indicate fragility	Moderate Tanimoto (0.63-0.68) needed for CFs	Both suggest the scaffold is **robust** but substituent is **fragile**

WHERE THEY DISAGREE:

Aspect	MolCE	Counterfactuals	Explanation
Mechanism	Identifies **specific R-groups** that could enhance activity	Shows **what breaks** activity	MolCE asks *"why active?"*; Counterfactuals asks *"what breaks it?"*
Top R-groups	`*CC(C)NC(C)=O`, `*C=C(F)C(=O)OCC`, `*C(O)CNC(C)CC`	`COCCCO-`, `COC(C)C(N)-`, `COC(=O)CO-`	MolCE proposes **alternative active substituents**; Counterfactuals shows **deactivating modifications**
Scaffold Analysis	Only 1 scaffold evaluated (contrast = 0.0)	No scaffold changes attempted	MolCE finds **scaffold is stable**; Counterfactuals focuses only on **substituent perturbations**

Key Insight:

> MolCE reveals **what substituents would enhance or maintain activity** (contrastive R-groups), while Counterfactuals shows **what modifications destroy activity**. They complement each other: MolCE guides **optimization** (what to keep/change for activity), Counterfactuals reveals **fragility** (what breaks activity). **The substituent position is universally identified as the critical determinant by both methods.**

📋 Overall Cross-Method Consensus

Structural Determinant	MolAnchor	SHAP	MolCE	Counterfactuals	Consensus
Diazepane Core	✅ Anchor	✅ Top bits	🟣 Scaffold stable	🟣 Not tested	**Critical scaffold**
Benzylamine Substituent	✅ Part of anchor	✅ Bit 1792 (5.74)	✅ Site 1 critical	✅ All CFs modify here	**Key activity determinant**
Thienyl Moiety	✅ Part of anchor	✅ Bit 885 (5.39)	🟣 Not isolated	🟣 Not isolated	**Important but complex**
Substituent Position	🟣 Implicit	🟣 Implicit	✅ **Explicit**	✅ **Explicit**	**Most critical**
Negative Features	❌ Not shown	✅ Bit 552 (-5.37)	🟣 Not shown	🟣 Not shown	**SHAP unique insight**

🎯 Final Conclusions

✅ Agreements Across All Methods:

1. **The diazepane core** is the essential scaffold for D(2) receptor binding
2. **The benzylamine substituent** is the primary activity determinant
3. **The substituent position** is the most sensitive/important feature

⚠️ Key Disagreements:

1. **Granularity:** MolAnchor (coarse) vs SHAP (fine-grained) — different levels of chemical insight
2. **Direction:** MolCE (constructive: what maintains/enhances activity) vs Counterfactuals (destructive: what breaks activity)
3. **Negative Features:** Only SHAP identifies **detrimental** molecular features (Bit 552: `ccs`)

💡 Recommendation for Lead Optimization:

- **Keep:** Diazepane core, benzylamine connectivity
- **Modify:** Substituent on diazepane nitrogen (position 1) — this is where MolCE and Counterfactuals converge
- **Avoid:** Certain thienyl configurations (SHAP negative bit)
- **Explore:** MolCE's proposed R-groups for activity enhancement

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*This benchmarking study reveals that while all four methods converge on the **substituent position** as the key determinant, they provide complementary insights: MolAnchor and SHAP explain **why** the compound is active, while MolCE and Counterfactuals guide **how** to modify it.* 🧬